

## Module 7: Redshift Assignment

Problem Statement: You work for XYZ Corporation. Their application requires a database service that can store data which can be retrieved if required.

Implement suitable service for the same. While migrating, you are asked to perform the following tasks:

1. Create a Redshift data warehouse.
2. Using the query editor:
  - a. Load some data
  - b. Query the data

### SOLUTION:

#### Task 1: Create a Redshift Data Warehouse

1. Log in to the AWS Management Console: Open the Amazon Redshift console at <https://console.aws.amazon.com/redshift/>.

2. Create Cluster:

Click on "Clusters" in the left sidebar.

Click on "Create cluster".

3. Configure your cluster settings:

4. Cluster identifier: Enter a unique name for your Redshift cluster.

5. Node type: Choose the appropriate node type based on your workload and performance requirements.

Number of nodes: Select the number of nodes for your cluster.

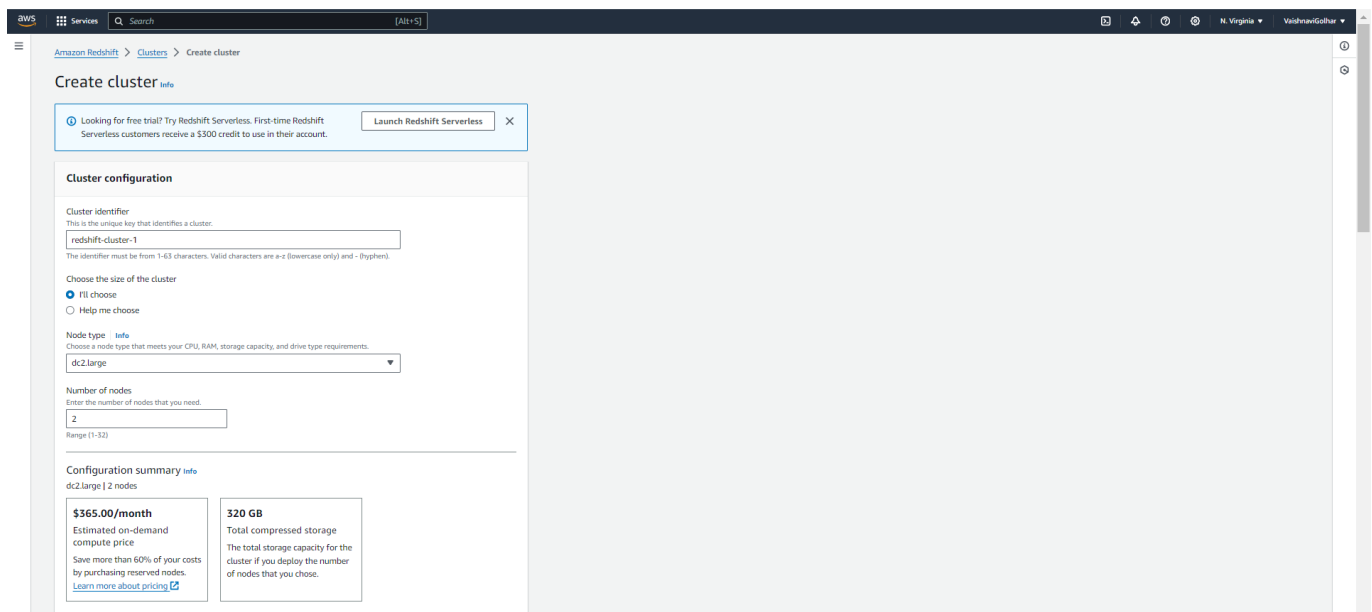
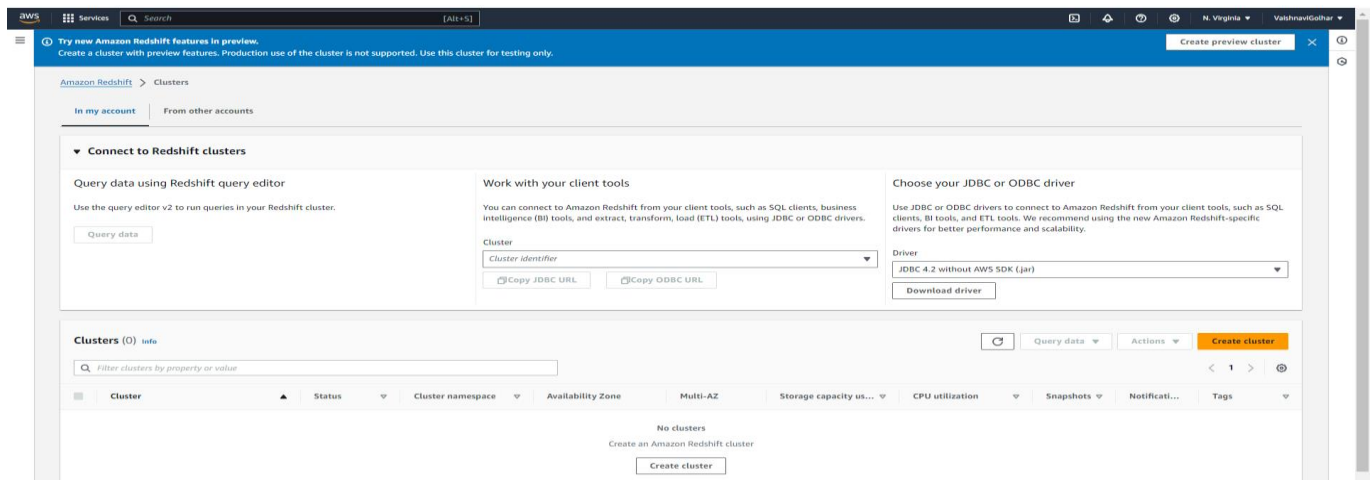
6. Database name: Enter the name of the initial database to be created in the cluster.

7. Master user name and password: Set the master user credentials for accessing the cluster.

8. VPC, subnet group, and security groups: Configure network settings for your cluster.

9. Click "Create cluster".

10. Wait for Cluster Creation: It may take some time for the cluster to be created. Monitor the status on the Clusters page until it shows as "Available".





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Cluster permissions

Create an IAM role as the default for this cluster that has the [AmazonRedshiftAllCommandsFullAccess](#) policy attached. This policy includes permissions to run SQL commands to COPY, UNLOAD, and query data with Amazon Redshift. The policy also grants permissions to run SELECT statements for related services, such as Amazon S3, Amazon CloudWatch logs, Amazon SageMaker, and AWS Glue.

Associated IAM roles (1/1)

Set defaultManage IAM roles

Create, associate, or remove an IAM role. You can associate up to 50 IAM roles. You can also choose an IAM role and set it as the default for this cluster.

Search for associated IAM role by name, status, or role type

☒

IAM roles

☐

Status

☐

Role type

|                                     |                                                   |             |         |
|-------------------------------------|---------------------------------------------------|-------------|---------|
| <input checked="" type="checkbox"/> | AmazonRedshift-CommandsAccessRole-20240323T120404 | Not applied | Default |
|-------------------------------------|---------------------------------------------------|-------------|---------|

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Identity and Access Management (IAM)

Dashboard

Access management

User groups

Users

**Roles**

Policies

Identity providers

Account settings

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies (SCPs)

Related consoles

[IAM Identity Center](#)

[AWS Organizations](#)

IAM > Roles > AmazonRedshift-CommandsAccessRole-20240323T120404

AmazonRedshift-CommandsAccessRole-20240323T120404

Delete

Edit

Summary

Creation date

March 23, 2024, 12:04 (UTC+05:30)

ARN

arn:aws:iam::161681788148:role/service-role/AmazonRedshift-CommandsAccessRole-20240323T120404

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Access Advisor

Revoke sessions

Permissions policies (2)

You can attach up to 10 managed policies.

Filter by Type

Search

All types

| <input type="checkbox"/> | Policy name                                                         | Type             | Attached entities |
|--------------------------|---------------------------------------------------------------------|------------------|-------------------|
| <input type="checkbox"/> | <a href="#">AmazonRedshift-CommandsAccessPolicy-20240323T120404</a> | Customer managed | 1                 |
| <input type="checkbox"/> | <a href="#">AmazonRedshiftAllCommandsFullAccess</a>                 | AWS managed      | 1                 |

Permissions boundary (not set)

Set a permissions boundary to control the maximum permissions this role can have. This is not a common setting but can be used to delegate permission management to others. [Learn more about permission boundaries](#)

## Additional configurations ☒ Use defaults

These configurations are optional, and default settings have been defined to help you get started with your cluster. Turn off "Use defaults" to modify these settings now.

### Network

Using default VPC (vpc-0b22f56f603e7fa1) and default subnet.

### Security

Using default (sg-08693198bc1703d9e) cluster security group.

### Configuration

Using default:redshift-1.0 parameter group with no database encryption.

### Backup

Automated snapshots are created about every eight hours or following every 5 GB per node of data changes, whichever comes first.

### Maintenance

Using current maintenance track.

Cancel

Create cluster

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Sample data load

After the cluster is created, Amazon Redshift starts to load the sample data.

Amazon Redshift is creating redshift-cluster-1.

Clusters (1) Info

Query data

Actions

Create cluster

Filter clusters by property or value

< 1 >

| <input type="checkbox"/> | Cluster                                           | Status   | Cluster namespace      | Availability Zone | Multi-AZ | Storage capacity us... | CPU utilization | Snapshots | Notificati... | Tags |
|--------------------------|---------------------------------------------------|----------|------------------------|-------------------|----------|------------------------|-----------------|-----------|---------------|------|
| <input type="checkbox"/> | redshift-cluster-1<br>dc2.large   1 node   160 GB | Creating | 0a068fed-1acc-479d-... | -                 | No       |                        |                 | -         |               |      |

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Try new Amazon Redshift features in preview.

Create a cluster with preview features. Production use of the cluster is not supported. Use this cluster for testing only.

Create preview cluster

Amazon Redshift > Clusters

In my account

From other accounts

Connect to Redshift clusters

Query data using Redshift query editor

Use the query editor v2 to run queries in your Redshift cluster.

Query data

Work with your client tools

You can connect to Amazon Redshift from your client tools, such as SQL clients, business intelligence (BI) tools, and extract, transform, load (ETL) tools, using JDBC or ODBC drivers.

Cluster

redshift-cluster-1

Copy JDBC URL

Copy ODBC URL

Choose your JDBC or ODBC driver

Use JDBC or ODBC drivers to connect to Amazon Redshift from your client tools, such as SQL clients, BI tools, and ETL tools. We recommend using the new Amazon Redshift-specific drivers for better performance and scalability.

Driver

JDBC 4.2 without AWS SDK (jar)

Download driver

Clusters (1) Info

Query data

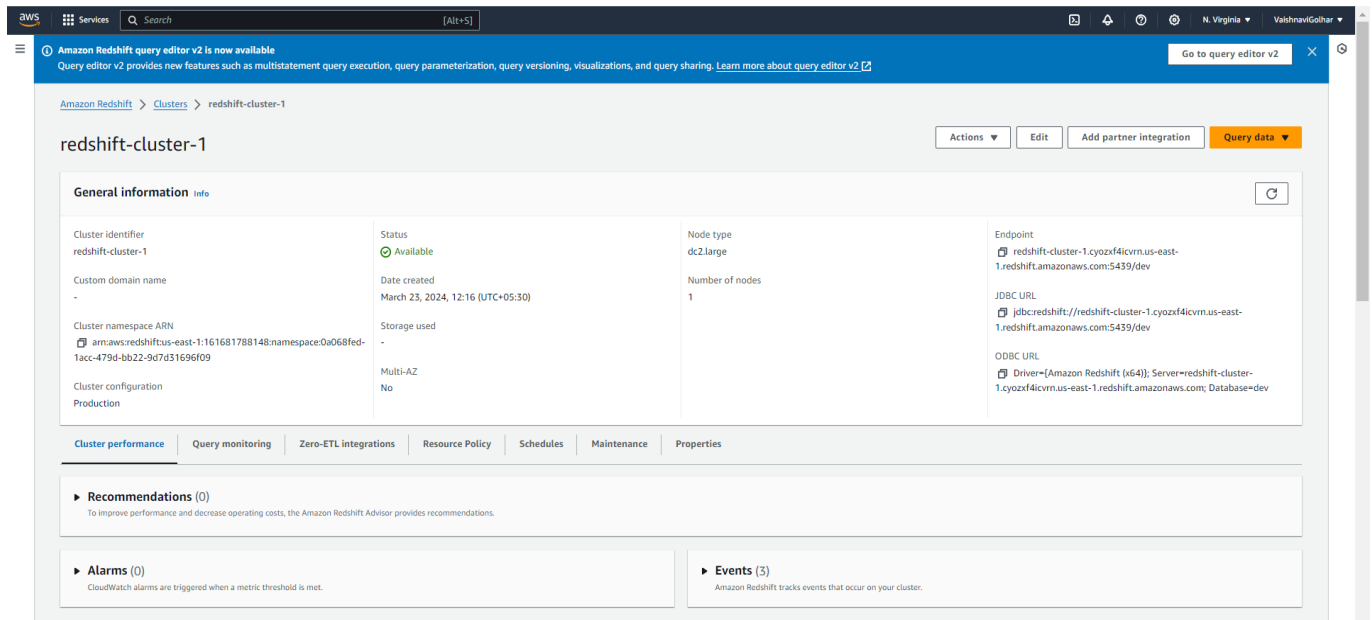
Actions

Create cluster

Filter clusters by property or value

< 1 >

| <input type="checkbox"/> | Cluster                                           | Status    | Cluster namespace      | Availability Zone | Multi-AZ | Storage capacity us... | CPU utilization | Snapshots | Notificati... | Tags |
|--------------------------|---------------------------------------------------|-----------|------------------------|-------------------|----------|------------------------|-----------------|-----------|---------------|------|
| <input type="checkbox"/> | redshift-cluster-1<br>dc2.large   1 node   160 GB | Available | 0a068fed-1acc-479d-... | us-east-1b        | No       |                        |                 | -         |               |      |



## Task 2: Using the Query Editor

### 1. Launch Query Editor:

Once your Redshift cluster is available, click on the cluster name to view its details.

### 2. Click on the "Query editor" tab.

Click on "Launch Query Editor".

### 3. Load Some Data:

In the Query Editor, write and execute SQL queries to load data into your Redshift database.

You can use commands like COPY to load data from Amazon S3, or manually insert data using INSERT INTO statements.

### 4. Query the Data:

Write SQL queries in the Query Editor to retrieve and analyze the data you loaded.

Amazon Redshift

Redshift Serverless

Provisioned clusters dashboard

Clusters

Query editor

Query editor v2

Queries and loads

Datashares

Zero-ETL Integrations

IAM Identity Center connections

Configurations

AWS Partner Integration

Informatica Data Loader

Advisor

Amazon Redshift > Query editor

Editor | Query history | Saved queries | Scheduled queries

Connect to a database to run queries and view results.

Resources

Select database

Select schema

Filter tables

No resources

Query 1

1

Connect and run

Save

Schedule

Clear

Send feedback

Query results

Table details

No resources

Connect to a database to run queries and view results.

Amazon Redshift

Redshift Serverless

Provisioned clusters dashboard

Clusters

Reserved nodes

Snapshots

Query editor

Query editor v2

Queries and loads

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Query editor v2 provides new features such as multistatement query execution, query parameterization, query versioning, visualizations, and query sharing. Learn more about query editor v2

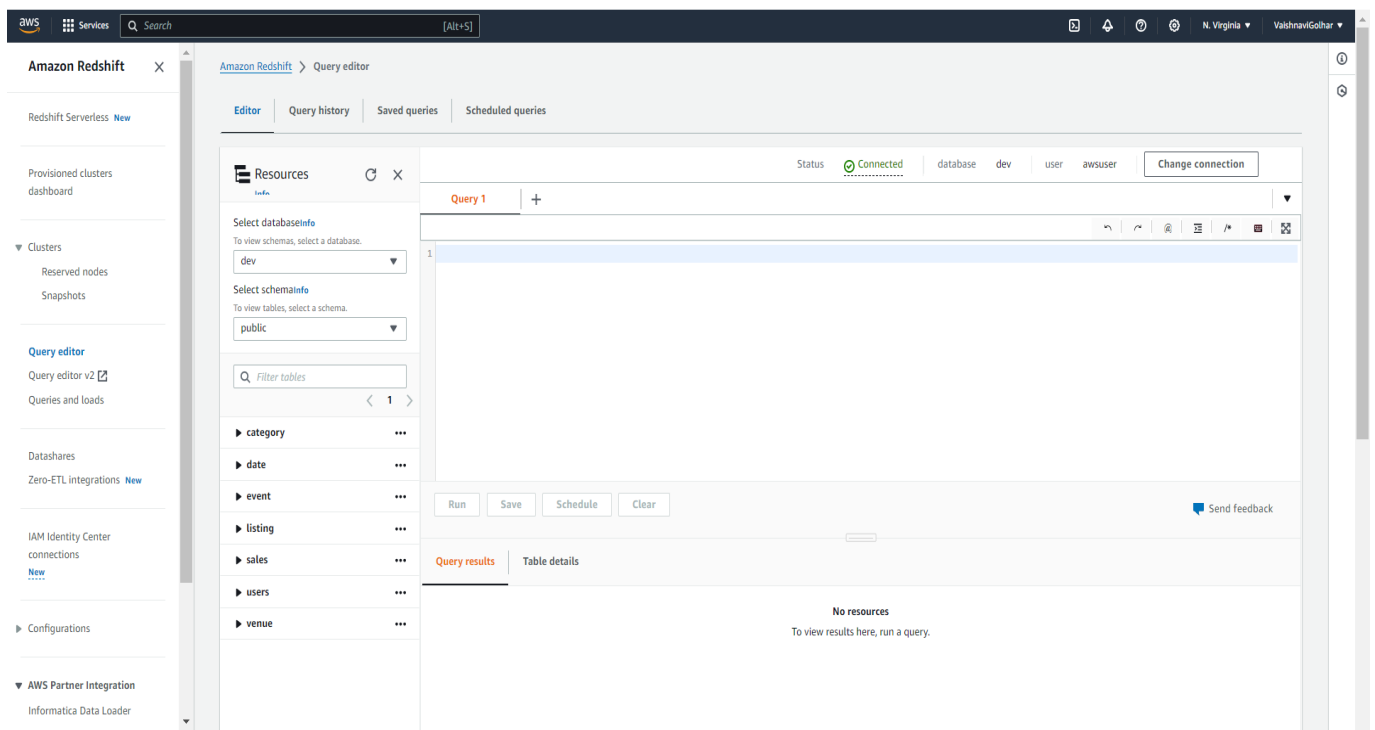
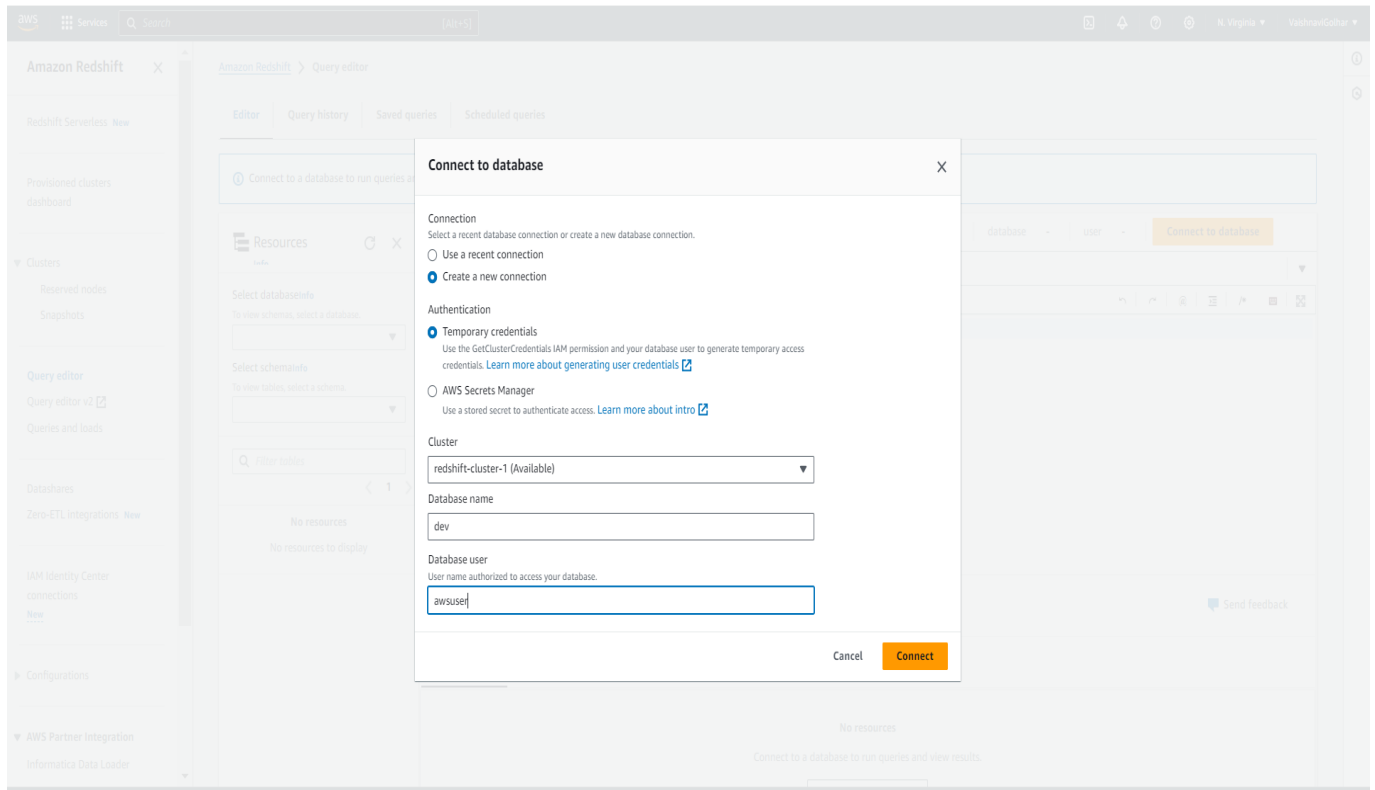
|                                                                                        |                                   |                 |                                                                                                                     |
|----------------------------------------------------------------------------------------|-----------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------|
| Custom domain name                                                                     | Date created                      | Number of nodes | JDBC URL                                                                                                            |
| -                                                                                      | March 23, 2024, 12:16 (UTC+05:30) | 1               | jdbc:redshift://redshift-cluster-1.cyozx4icvm.us-east-1.redshift.amazonaws.com:5439/dev                             |
| Cluster namespace ARN                                                                  | Storage used                      |                 | ODBC URL                                                                                                            |
| arn:aws:redshift:us-east-1:161681788148:namespace:0a068fed-1acc-479d-bb22-9d7d51696f09 | 0.22% (0.35 of 160 GB used)       |                 | Driver={Amazon Redshift (x64)}; Server=redshift-cluster-1.cyozx4icvm.us-east-1.redshift.amazonaws.com; Database=dev |
| Cluster configuration                                                                  | Multi-AZ                          |                 |                                                                                                                     |
| Production                                                                             | No                                |                 |                                                                                                                     |

Cluster performance | Query monitoring | Zero-ETL integrations | Resource Policy | Schedules | Maintenance | Properties

Database configurations

Edit admin credentials | Rotate encryption keys | Edit

|                 |                                                                                                                                                                                                                                                                                                                                                                                                          |                |               |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|
| Database name   | Parameter group                                                                                                                                                                                                                                                                                                                                                                                          | Encryption     | Audit logging |
| dev             | Defines database parameter and query queues for all the databases.                                                                                                                                                                                                                                                                                                                                       | Disabled       | Disabled      |
| Port            | SSH ingestion setting (cluster public key)                                                                                                                                                                                                                                                                                                                                                               | AWS KMS key ID |               |
| 5439            | ssh-rsa<br>AAAAB3NzaC1yc2EAAAADAQABAAQDQXKLnEAAZ3ZfoohjMzpJlOC7KcWUhaQBWQ48RH8i6dhsyYubzoOM9djhDGK9/gWxHxcMTs7PUf4q+HDWXS5GAPoH0378Y216P1YZ0e820kqgSWNeeyx1uulhfnqpD34lwRzrZ5pk2/JumHs6oYozMTbVpAex1K5BDYpROiKWGOrLaBBQ2wOkssm9v9uf9LAD9WYi6oJ4Kxb5puZSjlmzRSFRTIn5Tm93XMBQ94C3qCuMYh8KQEZv534P81qWScZ+C6fCVJRjyFmWxqjgXkUSKIZuUshWahZE305220iGPHYHnd0un1k/bPNCNeAmJoMnNOrOjipJg9B7tpRVZ Amazon-Redshift | -              |               |
| Admin user name |                                                                                                                                                                                                                                                                                                                                                                                                          |                |               |
| awsuser         |                                                                                                                                                                                                                                                                                                                                                                                                          |                |               |





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Amazon Redshift

Redshift Serverless **New**

Provisioned clusters dashboard

▼ Clusters

Reserved nodes

Snapshots

**Query editor**

Query editor v2

Queries and loads

Datashares

Zero-ETL integrations **New**

IAM Identity Center connections

**New**

Amazon Redshift > Query editor

Editor | Query history | Saved queries | Scheduled queries

Resources Info

Select databaseInfo

To view schemas, select a database.

dev

Select schemaInfo

To view tables, select a schema.

public

Filter tables

< 1 >

category

date

event

listing

sales

users

Status **Connected**

database dev user awsuser

Change connection

Query 1

+

1 create table shoes (shoetype varchar (255),

2 color varchar (11));

Run

Save

Schedule

Clear

Send feedback

Query results

Table details

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Redshift Serverless **New**

Provisioned clusters dashboard

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event

listing

sales

shoes

users

venue

Status **Connected**

database dev user awsuser

Change connection

Query 1

+

1 create table shoes (shoetype varchar (255),

2 color varchar (11));

Run

Save

Schedule

Clear

Send feedback

Query results

Table details

Query

Execution

Data

Visualize

Completed, started on March 23, 2024 at 12:29:39

ELAPSED TIME: 00 m 03 s

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Amazon Redshift

Redshift Serverless

Provisioned clusters dashboard

Clusters

- Reserved nodes
- Snapshots

Query editor

- Query editor v2
- Queries and loads

Datashares

Zero-ETL integrations

IAM Identity Center connections

Resources Info

Select database

dev

Select schema

public

Filter tables

category

date

event

listing

sales

shoes

users

venue

Status

Connected

database

dev

user

awsuser

Change connection

Query 1

1 insert into shoes values

2 ('loafers','brown');

3 ('sandles','black');

4

Run

Save

Schedule

Clear

Send feedback

Query results

Table details

Query 1816

Execution

Data

Visualize

Completed, started on March 23, 2024 at 12:33:11

ELAPSED TIME: 00 m 03 s

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Amazon Redshift

Query editor

Editor

Query history

Saved queries

Scheduled queries

Resources Info

Select database

dev

Select schema

public

Filter tables

category

date

event

listing

Status

Connected

database

dev

user

awsuser

Change connection

Query 1

1 select \* from shoes;

2

Run

Save

Schedule

Clear

Send feedback

aws

Services

Search

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public

Filter tables

< 1 >

category

date

event

listing

sales

shoes

users

venue

Run

Save

Schedule

Clear

Send feedback

Query results

Table details

Query 1870

Execution

Data

Visualize

Completed, started on March 23, 2024 at 12:34:31

ELAPSED TIME: 00 m 02 s

Rows returned (2)

Export

Search rows

< 1 >

| shoetype | color |
|----------|-------|
| loafers  | brown |
| sandles  | black |

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Services

Search

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Amazon Redshift

Query editor

Editor

Query history

Saved queries

Scheduled queries

Queries and loads (4)

Info

Open in Editor

Filter queries

< 1 >

|                          | Start time                                  | Status   | Duration | SQL                                                                |
|--------------------------|---------------------------------------------|----------|----------|--------------------------------------------------------------------|
| <input type="checkbox"/> | Mar 23rd, 2024 12:34:28 PM<br>2 minutes ago | Finished | 782 ms   | select * from shoes;                                               |
| <input type="checkbox"/> | Mar 23rd, 2024 12:34:17 PM<br>2 minutes ago | Failed   | 499 ms   | select * from shoes;                                               |
| <input type="checkbox"/> | Mar 23rd, 2024 12:33:08 PM<br>3 minutes ago | Finished | 879 ms   | insert into shoes values ('loafers','brown'), ('sandles','black'); |
| <input type="checkbox"/> | Mar 23rd, 2024 12:29:35 PM<br>6 minutes ago | Finished | 512 ms   | create table shoes (shoetype varchar (255), color varchar (11));   |